Subsidies for truthful cooperation: sharp proportional bounds and nonlinear escape

Maxwell Flanagan

October 2026

Working manuscript for correctness, priority, and significance review

Abstract

Fixed owners supply capacities to k complementary resource pools; a coalition's worth is the minimum of its pool totals. We study nonnegative payments that cover every coalition and are nondecreasing in each owner's own capacity report. An optional floor protects $1/k$ of each owner's maximum payoff in the original efficient core. We prove an all-profile total-payment bound of order $\log k/\log\log k$, independent of owner counts, using capped owner coordinates and the established absolute continuous Banzhaf functional. Every universal total-payment constant must be at least two. Within the floor-free class of payments proportional to capped capacities in each pool, we determine the sharp first-order worst-case factor $(1+o(1))\log k/\log\log k$, even when pool prices depend on the entire labeled profile. Nonlinear payments cover the complete linked-profile family used in that lower bound at factor two; this extends to two altered pools with binary reports. A different nonlinear rule handles reports in $[0,1]$ with equal pool sizes and a fixed public reference capacity when the number of altered pools is bounded. Thus the proportional lower bound does not establish unrestricted divergence. Complete proofs and exact verification scripts are provided. Whether an absolute budget factor works on all profiles remains unresolved, as does historical priority for the proposed results.

Keywords: weighted multi-glove game; capacity reporting; coalition coverage; subsidy; continuous Banzhaf index; proportional payments.

1 Introduction

Complementary production creates a tension between coalitional stability and incentives to reveal capacity. A minimum-total pool can receive the entire surplus in a stable allocation, but increasing an owner's reported capacity can change which pool receives it. An external subsidy permits payment rules that retain coalition coverage while relaxing efficiency.

This paper quantifies that tradeoff in weighted multi-glove games. Owners and pool memberships remain fixed. A player can underreport its own capacity, and its utility is its payment; there are no operating costs. We require a single payment rule across all report profiles, rather than independently choosing a core allocation at each profile. When requested, the rule also protects a fraction of an individual's best payoff in the *original* core. We never recompute this benchmark in a subsidized game.

Our central comparison is between proportional and nonlinear owner payments. A capped proportional rule achieves a sublogarithmic budget factor, and an explicit sequence of report changes proves that its first-order growth is optimal throughout the proportional class. The same report sequences admit bounded nonlinear payments. This supplies a precise limitation of the lower bound and motivates the remaining unrestricted question.

Results and their domains. Let T denote the grand coalition's worth, B the total-payment factor, and $\beta = B - 1$ the subsidy factor. These quantities are distinct.

- On all nonnegative real profiles, with arbitrary finite owner counts, the capped Banzhaf base has factor at most $R_k + 1/\log R_k$, where $R_k > e$ solves $R_k \log(R_k/e) = \log k$. The original-core floor adds at most one to that factor (Theorem 3.1).
- Every constant B uniform in both pool and owner counts satisfies $B \geq 2$, even without the floor (Theorem 4.1).
- Floor-free capped-proportional rules have optimal first-order factor $(1 + o(1)) \log k / \log\log k$ (Corollary 5.2). This is a class minimax statement, not the unrestricted optimum.
- On the binary linked-profile family, nonlinear payments have factor 2, or 3 with the floor (Theorem 6.1). The factor survives two simultaneously altered pools (Appendix A).
- With r owners per pool, reports in $[1/r, 1]$, and at most m pools altered from all-unit, a different rule has base factor $2m + 3$ and fair factor $2m + 4$ (Theorem 7.1). Its constants are uniform in k, r and the report grid when m is fixed.

The last two results are restricted-domain feasibility theorems. They do not assert that a bounded nonlinear rule exists on all report profiles.

1.1 Established ingredients and proposed contribution

Moretti and Norde [1, Definition 3.1, Theorem 3.2] define the weighted multi-glove model and prove total balancedness. Their final remark already extends the results to positive real weights and discusses zeros. We give the minimum-pool core construction directly for nonnegative capacities. Huang et al. [2] study binding capacity underreports without use costs in max-flow games and prove that efficient core selection need not be truthful. Our game is the specialization to serial bundles of parallel edges. We relax efficiency and retain coverage of the original coalition values.

The raw integral functional in our upper rule is the absolute continuous Banzhaf index of Kurz [3, Definition 6.11, arXiv v1, p. 17]. We prove the exact identification in Section 3; we do not claim to introduce that index. The proposed contributions are the joint report-monotonicity, coverage and budget guarantees after capping, the quantitative unrestricted lower family, the sharp proportional-class asymptotic, and the nonlinear feasible assignments that delimit the obstruction.

Cost-of-stability work [4] studies subsidies for stabilizing a fixed game. Here every fixed game already has a nonempty core; the positive budget requirement arises from linking its allocations to other report profiles. Population or cross-monotonic sharing schemes, including the max-flow setting in Immorlica et al. [5], impose comparisons across participating-player sets. Those comparisons are not assumed here. A capacity unit is not a separately strategic clone of its owner.

These distinctions identify the claim being submitted for review. They do not certify priority. The accompanying source ledger records the original passages inspected and the remaining literature questions.

2 Model and two reductions

Let $N = N_1 \sqcup \cdots \sqcup N_k$ be a fixed finite partition with $k \geq 2$ and nonempty pools. Reports are $c \in \mathbb{R}_+^N$. Write

$$C_j = \sum_{i \in N_j} c_i, \qquad v_c(S) = \min_j \sum_{i \in S \cap N_j} c_i, \qquad T = \min_j C_j.$$

A payment rule $y(c) \in \mathbb{R}^N_+$ *covers coalitions* if $y(c)(S) \geq v_c(S)$ for every $S \subseteq N$. It is *own-report monotone* if $y_i(t, c_{-i})$ is nondecreasing in t. Under binding underreports and payment utility, this is exactly the required no-profitable- underreport condition. We impose no overreport, group-report, split, merge, or population-change guarantee. On a restricted domain, all comparisons whose endpoints belong to that domain are required.

The original efficient core and coordinate benchmarks are

$$\mathcal{C}(c) = \{x : x(N) = T,\ x(S) \geq v_c(S)\ \forall S \subseteq N\}, \qquad H_i(c) = \max_{x \in \mathcal{C}(c)} x_i.$$

Paying capacities in one minimum-total pool and zero elsewhere gives a core allocation. Singleton constraints imply $x_i \geq 0$, so this nonempty core is compact. Our optional fairness requirement is $y_i(c) \geq H_i(c)/k$. A factor B means $y(c)(N) \leq BT$ at every profile in the stated domain. At $T = 0$ we set payments to zero.

Lemma 2.1 (Marginal floor)**.** *Put $M_i = T - v_c(N \setminus \{i\})$. Then $H_i \leq M_i$, M_i is own-report monotone, and $\sum_i M_i \leq kT$. If a nonnegative covering own-monotone base has factor B, replacing y_i by $\max\{y_i, M_i/k\}$ gives the original floor with factor at most $B + 1$.*

Proof. Core efficiency and the complementary coalition constraint give $x_i = T - x(N \setminus \{i\}) \leq M_i$. If $i \in N_j$, set $A_i = C_j - c_i$ and $D_j = \min_{\ell \neq j} C_\ell$. Then

$$M_i = \min\{c_i, (D_j - A_i)_+\} = (c_i - (C_j - T))_+.$$

The first form proves own-monotonicity. For $s = C_j - T \geq 0$, if $p \geq 1$ terms $(c_i - s)_+$ in pool j are positive, their sum is at most $C_j - ps \leq C_j - s = T$; if none is positive it is zero. Sum over pools. The maximum of two own-monotone functions is own-monotone, and its additional cost is at most $\sum_i M_i/k \leq T$. □

Consequently the existence of a universal constant factor with this floor is equivalent to its existence without the floor, although the best constants need not coincide.

Lemma 2.2 (Capping)**.** *Let $d_i = \min\{c_i, T\}$ and $D_j = \sum_{i \in N_j} d_i$. Then*

$$v_c(S) = \min_j \sum_{i \in S \cap N_j} d_i, \qquad \min_j D_j = T.$$

Proof. For any owner set A, $\min\{c(A), T\} = \min\{d(A), T\}$: if it contains an owner above T, both truncated sums are T; otherwise the sums agree. Coalition values are at most T. A minimum-total original pool has no capacity above T, and has capped total T. Taking minima proves the claim. □

3 An all-profile upper bound

For $T > 0$, draw independent $U_i \sim \text{Unif}[0, 1]$ as an integration device and define

$$X_j = \sum_{i \in N_j} d_i U_i, \qquad p_j = \mathbb{P}(X_j < X_\ell \text{ for all } \ell \neq j), \qquad z_i = d_i p_j \quad (i \in N_j).$$

Every pool has a positive capped capacity. Its sum X_j is continuous, and the pool sums are independent. Thus ties have probability zero and $\sum_j p_j = 1$. The mechanism pays deterministic expectations, not random realizations of these amounts. Let $y_i = \max\{z_i, M_i/k\}$.

Theorem 3.1. *The base z covers every coalition and is own-report monotone on all nonnegative real profiles. For $R_k > e$ defined by $R_k \log(R_k/e) = \log k$,*

$$\sum_i z_i \le \left(R_k + \frac{1}{\log R_k}\right) T, \qquad \sum_i y_i \le \left(1 + R_k + \frac{1}{\log R_k}\right) T.$$

Both factors are $(1 + o(1)) \log k / \log\log k$ as $k \to \infty$, uniformly over all finite owner counts. The second rule meets H_i/k.

Proof. By Lemma 2.2, for every S,

$$z(S) = \sum_j p_j d(S \cap N_j) \ge \min_j d(S \cap N_j) = v_c(S).$$

For own-monotonicity, fix $i \in N_j$, write its report as x, let $a = \sum_{h \in N_j \setminus \{i\}} c_h$ and $b = \min_{\ell \ne j} C_\ell$. If $b = 0$ all payments vanish. Otherwise $T(x) = \min\{x + a, b\}$ and $d_i(x) = \min\{x, b\}$. Conditional on U_{-i}, put

$$R = \sum_{h \in N_j \setminus \{i\}} U_h \min\{c_h, T(x)\}, \qquad Y(x) = \min_{\ell \ne j} \sum_{h \in N_\ell} U_h \min\{c_h, T(x)\}.$$

Here R is independent of x: if $a \le b$, every other own-pool capacity is at most $a \le T(x)$; if $a > b$, $T(x) = b$ throughout. Also $Y(x)$ is nondecreasing. Integrating only U_i gives

$$d_i \mathbb{P}(U_i d_i < Y - R \mid U_{-i}) = \min\{d_i, (Y - R)_+\}.$$

At $d_i = 0$ both sides are zero. The expression is nondecreasing in x; its expectation is z_i. Lemma 2.1 proves the floor assertions.

For the budget, normalize $a_i = d_i/T \in [0, 1]$, put $\rho_j = D_j/T$, and let W be the winning pool. Then $z(N)/T = \mathbb{E}\rho_W$. A short pool has $X_s/T \le 1$, so $p_j \le \mathbb{P}(\sum_{i \in N_j} a_i U_i \le 1)$. For $t > 0$ and $a \in [0, 1]$, concavity of $u \mapsto u^a$ gives

$$\mathbb{E}e^{-taU} \le (\mathbb{E}e^{-tU})^a = \left(\frac{1 - e^{-t}}{t}\right)^a \le t^{-a}.$$

Independence and exponential Markov imply $p_j \le e^t t^{-\rho_j}$. Set $t = \rho_j$. Because $(e/s)^s$ decreases for $s \ge e$,

$$\mathbb{P}(\rho_W > s) \le \min\{1, k(e/s)^s\}, \qquad s \ge e.$$

At $s = R_k$ the second expression is one; its logarithmic derivative is $-\log s \le -\log R_k$ thereafter. Tail integration therefore yields

$$\mathbb{E}\rho_W \le R_k + \int_{R_k}^{\infty} e^{-(s - R_k)\log R_k}\, ds = R_k + \frac{1}{\log R_k}.$$

Finally, $R_k \log(R_k/e) = \log k$ implies $R_k \sim \log k / \log\log k$. Add the floor cost from Lemma 2.1. □

Identification with an established index. For a fixed profile define the continuous monotone game

$$f_d(u) = \frac{1}{T} \min_j \sum_{i \in N_j} d_i u_i, \qquad u \in [0, 1]^N.$$

It has endpoint values zero and one. Kurz's absolute continuous Banzhaf index integrates $f_d(1, u_{-i}) - f_d(0, u_{-i})$ over the other coordinates. With R, Y as above,

$$T\big[f_d(1, u_{-i}) - f_d(0, u_{-i})\big] = \min\{R + d_i, Y\} - \min\{R, Y\} = \min\{d_i, (Y - R)_+\}.$$

Hence $z_i = T\,\mathrm{BZI}_i(f_d)$. This pointwise identity does not by itself prove monotonicity across reports, since f_d changes with c; the preceding conditional integration argument supplies that step. Exact evaluation can require exponentially many subset-sum knots. No polynomial-time or arbitrary rounded-payment implementation is claimed.

4 An unrestricted necessary factor of two

The next theorem uses neither proportionality nor a fairness floor.

Theorem 4.1. *Fix $k \geq 3$ and $r \geq 2$ owners per pool, and set $t = 2r - 2$. Every nonnegative covering own-monotone rule with factor $1 + \beta$ on all real profiles satisfies*

$$\beta \geq \frac{(k-1)(t-1)}{t(k-1+2t)}.$$

Therefore any total-payment constant uniform over both pool and owner counts must be at least two.

Proof. Average the rule over all pool permutations and all within-pool owner permutations. Relabeling and averaging preserve the guarantees, so a bound for the resulting invariant rule bounds the original rule. This is a proof device on fixed owners, not permission for an owner to split or merge. Put $q = r - 2$, so $t = 2q + 2$. Use two pool types

$$L = (1, 1, \underbrace{2, \ldots, 2}_{q}), \qquad H = (1, \underbrace{2, \ldots, 2}_{q}, t).$$

Their totals are t and $2t - 1$, respectively. All capacities are positive. At profile X with one L and $k - 1$ copies of H, let the payments be $a, a, b, \ldots, b$ in L and $u, v, \ldots, v, w$ in each H. Write $P = 2a + qb$, $Q = u + qv + w$. The budget is $P + (k-1)Q \leq t(1 + \beta)$. Actual coalitions give

$$\begin{aligned} a + (k-1)u &\geq 1, & 2a + (k-1)v &\geq 2, \\ a + qb + (k-1)(u + qv) &\geq t - 1, & P + (k-1)w &\geq t. \end{aligned}$$

The inequality involving v is used only if $q > 0$. The choices in the L pool are, respectively, one unit owner, both unit owners, one unit plus all 2-owners, and all owners. Each H contributes its unit owner, one 2-owner, the unit plus all 2-owners, or its t-owner. Combine these inequalities with weights $1, q, t - 1, 1$. The left side is $t[P + (k-1)Q] - (t-1)(k-1)w$ and the right side is t^2. Hence

$$(t-1)(k-1)w \leq t^2\beta. \tag{1}$$

At profile Y with two L pools and $k - 2$ copies of H, denote analogous payments by A, B, U, V, W. Its budget is $2(2A + qB) + (k-2)(U + qV + W) \leq t(1 + \beta)$. Coalitions selecting one unit, both units, or all owners from each L yield

$$2A + (k-2)U \geq 1, \quad 4A + (k-2)V \geq 2, \quad 2(2A + qB) + (k-2)W \geq t.$$

Again omit the middle inequality if $q = 0$. Their combination with weights $1, q, 1$ is the budget sum plus $2(t-1)A \geq 2t - 1$, so

$$2(t-1)A \geq t - 1 - t\beta. \tag{2}$$

Raising one unit report in an L pool of Y from 1 to t gives X up to relabeling. Own-monotonicity gives $w \geq A$. Equations (1)–(2) therefore imply

$$(k-1)(t-1-t\beta) \leq 2(t-1)(k-1)A \leq 2(t-1)(k-1)w \leq 2t^2\beta,$$

which rearranges to the claim. No sign assumption on $t-1-t\beta$ is needed. For $r = n+1$, $t = 2n$, $k = 8n^2+1$, the displayed lower bound is $(2n-1)/(2n+1) \to 1$. □

The limit proves a necessary constant; it proves neither that a factor-two rule exists nor that the unrestricted optimum diverges.

5 Sharp asymptotics for proportional payments

Consider final base payments of the form

$$z_i(c) = d_i(c)\pi_j(c), \qquad i \in N_j,$$

where π_j is an arbitrary nonnegative function of the whole labeled profile and fixed ownership. There is no normalization, anonymity, continuity, locality, or independence assumption on these prices.

Theorem 5.1. *On $k \geq 2$ pools with $r \geq 2$ owners each, every covering own-monotone capped-proportional rule with factor B satisfies*

$$B \geq \frac{kr}{k+(r-1)r^r}. \tag{3}$$

The same lower bound holds for raw-capacity-proportional payments $c_i\pi_j$.

Proof. Let H be the all-unit profile. Its value is r, and its budget implies $\sum_j \pi_j(H) \leq B$. Let L_j have all owners of pool j at $1/r$ and all other owners at 1. It has value one. The coalition containing all of pool j and one owner of each other pool has worth one, so

$$\pi_j(L_j) + \sum_{\ell \neq j} \pi_\ell(L_j) \geq 1, \qquad \pi_j(L_j) + r\sum_{\ell \neq j} \pi_\ell(L_j) \leq B.$$

Multiply the first by r and subtract the second:

$$B + (r-1)\pi_j(L_j) \geq r. \tag{4}$$

Raise each of pool j's reports successively from $1/r$ to 1, obtaining $P_{j,0} = L_j, \ldots, P_{j,r} = H$. Every capacity is at most one and every profile value is at least one, so capping changes no report. At step h, own-monotonicity of the changing owner gives

$$\pi_j(P_{j,h+1}) \geq \pi_j(P_{j,h})/r, \qquad \text{hence} \qquad r^r\pi_j(H) \geq \pi_j(L_j).$$

This transports a *common pool price* between changes made by different owners; that identity is precisely where proportionality is used. Substitute in (4), sum over j, and use the budget at H:

$$kr \leq kB + (r-1)r^r \sum_j \pi_j(H) \leq [k+(r-1)r^r]B.$$

The uncapped version has the same proof because the witness capacities are never capped. □

A nonnegative inequality certificate. Write all constraints as left side at least right side. For each j, use r times its L_j coverage row, once its budget row $B - \pi_j - r\sum_{\ell\neq j}\pi_\ell \geq 0$, and for each $h = 0, \ldots, r-1$ use $(r-1)r^{h+1}$ times the actual monotonicity row $\pi_j(P_{j,h+1}) - \pi_j(P_{j,h})/r \geq 0$. Add once in total $(r-1)r^r$ times $B - \sum_j \pi_j(H) \geq 0$. Prices cancel, leaving (3). All weights are nonnegative integers.

Corollary 5.2. *Let A_k be the infimum of worst-case total-payment factors among families of floor-free capped-proportional rules, defined for every finite fixed ownership with k pools and all nonnegative real capacities, covering all coalitions and satisfying own-monotonicity. Explicitly, if $\mathfrak{P}_k$ contains all finite ownership partitions with k nonempty pools and $\mathcal{F} = (z^P)_{P\in\mathfrak{P}_k}$ ranges over such rule families, then*

$$A_k = \inf_{\mathcal{F}} \sup_{P\in\mathfrak{P}_k} \sup_{c:T(c)>0} \frac{z^P(c)(N)}{T(c)}.$$

Rules for different fixed ownerships need not satisfy cross-ownership consistency. We have

$$A_k = (1+o(1))\frac{\log k}{\log\log k}.$$

Proof. For fixed $\epsilon > 0$, let $L = \log k$ and $r = \lfloor (1-\epsilon)L/\log L \rfloor$. For all sufficiently large k, $r \geq 2$ and $\log((r-1)r^r) \leq (1-\epsilon)L + o(L)$. Thus $(r-1)r^r/k \to 0$, and Theorem 5.1 gives $A_k \geq (1-o(1))r$. Take the lower limit and then $\epsilon \downarrow 0$. The base of Theorem 3.1 belongs to this class and gives the matching upper limit. □

The floor-restoring maximum need not remain proportional. Accordingly the corollary is floor-free; it is not a claim about the minimax among final proportional rules also meeting the floor, nor about unrestricted rules.

6 A nonlinear escape on the linked profiles

Fix equal pool size r. Let $\mathcal{D}_1$ contain *every labeled* profile with reports in $\{1/r, 1\}$ and at most one pool containing a low report. The family includes all orders of the report changes in Theorem 5.1.

Theorem 6.1. *On $\mathcal{D}_1$ there are nonnegative covering payments that are own-monotone on every retained report edge and have factor at most 2. The original-core floor can be added with factor at most 3.*

Proof. At H, pay $1/k$ to each owner. Otherwise let the altered pool have h unit owners, $0 \leq h < r$, and define

$$T_h = h + (r-h)/r, \quad \delta_h = (r-h)/r, \quad z_h = T_h/r,$$
$$a_h = \frac{(\delta_h - z_h)_+}{r-h}, \qquad b_h = 1 - z_h, \qquad e_h = \frac{z_h}{k-1}. \tag{5}$$

Pay its low owners a_h, its unit owners b_h, and every outside owner e_h. For a coalition, let b be its unit-owner count in the altered pool, s its selected low capacity, and n the smallest selected-owner count outside that pool. Its worth is $\min\{b+s, n\}$, and its payment is at least

$$b(1-z_h) + s(1 - z_h/\delta_h)_+ + nz_h.$$

If $n \leq b$, the first and last terms already give at least n. Otherwise $n \geq b+1$ because n, b are integers. It suffices that $z_h + s(1-z_h/\delta_h)_+ \geq s$, which follows from $s \leq \delta_h$ in both cases $z_h \leq \delta_h$ and $z_h \geq \delta_h$. At H, the average selected-owner count bounds its minimum, proving coverage there too.

The total off H is $F_h = h(1-z_h)+(\delta_h-z_h)_+ + rz_h$. If $z_h \le \delta_h$, this equals $T_h+(r-h-1)z_h \le 2T_h$. Otherwise it equals $h+(r-h)z_h \le T_h + T_h$.

Every retained increase raises a low owner to one. For $0 \le s \le r-2$, direct substitution in (5) gives

$$\frac{a_s}{b_{s+1}} = \frac{1}{r-1}\left(1 - \frac{s^2}{(r-s)(r-s-1)}\right)_+ \le 1, \tag{6}$$

with $b_{s+1} > 0$. Also $a_{r-1} = 0$. Thus $a_s \le b_{s+1}$ for internal edges and $a_{r-1} \le 1/k$ on the last-low boundary. This exhausts all labeled retained edges. Apply Lemma 2.1. □

The preceding certificate cannot show unrestricted divergence on these same profiles, even after adding the stated fairness floor. This is a separation *on the witness family*; it is not a separation between two global minimax asymptotics. Appendix A extends the factor-two base to two altered binary pools while preserving this boundary.

7 Continuous reports with a bounded number of altered pools

Here equal pool size $r \ge 2$ and the fixed public reference capacity 1 are assumptions. A pool is altered if at least one report differs from 1. There is no data-dependent choice of reference or ownership.

Theorem 7.1. *For reports in $[0,1]$ and at most m altered pools, the following rule is nonnegative, covering and own-monotone, with factor at most $2m+4$:*

$$b = \frac{1}{k}\min\left\{1, \frac{2T}{r-1}\right\}, \quad E_i = \sum_{h \in N_j \setminus \{i\}} (1-c_h), \quad y_i = b + \frac{2E_i}{r-1}\frac{Tc_i}{C_j} \quad (i \in N_j).$$

Set all payments to zero at $T=0$. On $[1/r, 1]$, the factor improves to $2m+3$. The original-core floor adds at most one in either domain.

Proof. When $T > 0$, $C_j \ge T$ so all denominators are positive. Fixing other reports makes E_i constant and nonnegative. If their own-pool sum is A_i, both $T(t)$ and $t/(A_i+t)$ are nonnegative and nondecreasing wherever the ratio is defined. Their product is nondecreasing; so is $b(T(t))$. At a zero-value endpoint all payments are zero. This proves own-monotonicity on the full cube, including all boundary transitions. If the changing owner was its pool's last non-unit owner, $E_i = 0$ throughout that change.

If $T \ge (r-1)/2$, then $b = 1/k$ and, writing $n_j = |S \cap N_j|$, the uniform payments give $|S|/k \ge \min_j n_j \ge v_c(S)$ because capacities are at most one. If $0 < T \le (r-1)/2$, choose a pool with $C_j = T$. For each of its owners, $E_i = r-1-(T-c_i) \ge (r-1)/2$, so its additional payment is at least c_i. Payments in this one pool alone cover every coalition. These two regimes include their common boundary; $T = 0$ is immediate.

Because $0 \le E_i \le r-1$, the additional payments in any pool sum to at most $2T$. They vanish in unaltered pools. The uniform part costs $r\min\{1, 2T/(r-1)\} \le 4T$. This proves the first bound. On $[1/r, 1]$ we have $T \ge 1$: for $r = 2$ the uniform part is at most $2 \le 2T$; for $r \ge 3$ it is at most $2rT/(r-1) \le 3T$. Lemma 2.1 proves the floor claim. □

For two altered pools this gives base $7T$ and fair $8T$ on $[1/r, 1]$. It revises the earlier binary payments: at $k = r = 2$, on $((1/2,1/2),(1,1))$, the new base is $(1,1,1/2,1/2)$, whereas Theorem 6.1 gives $(1/4,1/4,1/2,1/2)$. Preservation of the old boundary on continuous reports is not proved impossible, and none of these constants is asserted to be minimax sharp.

The formula is continuous when $T > 0$; at $T = 0$ its full-cube bound with $m = k$ forces every coordinate to tend to zero. Thus a single direct continuous rule works on every finite report grid. No inference from grid feasibility or exchange of limiting operations is required.

An exact limitation of this formula. Take $r = 3$, any $k \geq 2$, and $c_i = 1/3$ everywhere. Then $T = 1$, $E_i = 4/3$, $b = 1/k$, and $y_i = 1/k + 4/9$. Total payment is exactly $3 + 4k/3$. The floor is inactive since $M_i/k = 1/(3k) < b$. Therefore this particular rule has genuinely growing cost when all pools are altered. This says nothing about the best possible unrestricted rule.

8 What remains open

The unrestricted question is whether an absolute $B < \infty$ works for all k, all finite fixed pool sizes, and every nonnegative real profile. Our necessary factor two and general upper factor of order $\log k / \log \log k$ leave that question unresolved. The floor does not change constant existence by Lemma 2.1.

Any constant-budget construction must leave the capped-proportional class. The class lower proof cannot simply be relabeled as an unrestricted lower proof: a common price lets it transport constraints between different owners, and Theorem 6.1 supplies a feasible obstruction to that transfer. Theorem 7.1 similarly rules out divergent certificates supported solely on a fixed number of altered pools in its normalized, equal-size domain. These exclusions do not select between global feasibility and global divergence.

Three independent review questions remain: whether earlier work implies the joint guarantees or sharp class bound; whether a general nonlinear construction or unrestricted growing certificate is accessible; and whether this collection has significance beyond a specialized model. We make no claim to have resolved a recognized major open problem, to have established data-center applicability, or to have obtained human specialist endorsement.

Verification and assistance. The accompanying package contains standard-library exact-arithmetic programs, source hashes and reproducible commands. Tests check finite profiles and certificates; the universal and asymptotic claims rest on the written proofs. AI systems assisted the research, drafting and internal review. Independent model reviews are recorded as such and do not substitute for specialist review. The manuscript is a review draft, with historical novelty expressly unsettled.

A Two simultaneously altered binary pools

Use (5) and put $F_h = hb_h + (r - h)a_h + rz_h$. For an altered pool j with h unit owners, let R^j pay a_h, b_h to its own low and unit owners and e_h to *every* outside owner, including outside low owners. This vector covers the actual two-altered-pool game: set every outside report to one, apply Theorem 6.1, and observe that actual coalition values only decrease.

For $0 \leq s \leq r - 2$, let $q_s = a_s/b_{s+1}$. Equation (6) gives $0 \leq q_s \leq 1/(r - 1)$ and $b_{s+1} \geq (r - 1)a_s$. For $0 \leq g < h < r$ define

$$w(h, g) = \frac{1}{2} \prod_{s=g}^{h-1} q_s.$$

A zero factor gives weight zero; no reciprocal of q_s is used. For two altered pools with equal counts, use their equal mixture. With counts $h > g$, give the h-pool component weight $w(h, g)$ and the

g-pool component the remaining weight. With zero or one altered pool use the previous rule.

Theorem A.1. *This assignment covers all coalitions, is own-monotone on every retained binary edge, preserves Theorem 6.1 on its entire boundary, and has base factor at most* 2*, or factor* 3 *with the floor.*

Proof. Coverage follows from the component argument and convexity. Check the changing owner's payment on each possible increase.

- Equal counts s, s become $s+1, s$: the old payment is $(a_s + e_s)/2$, and the new one is $(q_s/2)b_{s+1} + (1 - q_s/2)e_s = a_s/2 + (1 - q_s/2)e_s$, which is larger.
- An already larger count $h > g$ increases internally: writing $w = w(h, g)$ and $w' = wq_h \le w$, old and new payments are $wa_h + (1-w)e_g$ and $w'b_{h+1} + (1-w')e_g = wa_h + (1-w')e_g$, respectively.
- A smaller count $g < h$ increases: put $w' = w(h, g+1)$, or $1/2$ at a tie, so $w = w'q_g \le w' \le 1/2$. The old payment is $(1-w)a_g + we_h$ and the new one is $(1-w')b_{g+1} + w'e_h$. For $r \ge 3$, $b_{g+1} \ge 2a_g$ makes the new own-component term at least a_g, while the outside term increases. For $r = 2$, the only case is $g = 0, h = 1$, where $q_0 = 1$ and $b_1 = a_0 = 1/4$, giving equality.
- The last low owner reaches one: $a_{r-1} = 0$, so its old payment is at most the other component's outside payment, which it receives exactly after its pool becomes unaltered. With no other altered pool its payment instead increases from zero to $1/k$.

One-altered internal edges were proved in Theorem 6.1.

For the budget we use, for integers $0 \le g < h < r$,

$$2T_g - F_g \ge 1/r, \quad F_h - F_g \le 2(h-g), \quad a_g > 0 \Longrightarrow F_g = T_g(2 - (g+1)/r). \tag{7}$$

Here is a direct verification. Extend h to real $x \in [0, r)$ in the formulas. Where $\delta_x \ge z_x$, $F(x) = T_x(2 - (x+1)/r)$ and $F'(x) \le 2$; its slack is $T_x(x+1)/r \ge 1/r$. Where $\delta_x \le z_x$, $F(x) = x + (r-x)T_x/r$ and $F'(x) = 2(r-1)(1-x/r)/r \le 2$. The formulas agree at the transition. At integer g in the latter region, $2T_g - F_g = (r-g)/r + gT_g/r \ge 1/r$. This proves (7).

Equal counts cost $F_g \le 2T_g$. Otherwise actual $T = T_g$, and total cost is $F_g + w(h, g)(F_h - F_g)$. A nonpositive difference is harmless. Set $d = h - g$. The excess is at most $d/(r-1)^d$ by (7) and the product bound. If $r \ge 4, d \ge 2$, this is at most $2/(r-1)^2 \le 1/r$, within the slack. If $d = 1, g \ge 1$, either $a_g = 0$ and the weight is zero, or the slack is $T_g(g+1)/r \ge 2/r$ and the excess is at most $1/(r-1) \le 2/r$. If $d = 1, g = 0, r \ge 3$, then

$$w = \frac{1}{2(r-1)}, \quad F_0 = 2 - 1/r, \quad F_1 - F_0 = \frac{(2r-1)(r-2)}{r^2},$$

and the excess is at most $1/r$ since $(2r-1)(r-2) \le 2r(r-1)$. The two remaining cases are $r = 2, h = 1, g = 0$, with cost $13/8 \le 2$, and $r = 3, h = 2, g = 0$, with cost $5/3 + (1/16)(10/9) = 125/72 \le 2$. These exhaust the parameters. Apply Lemma 2.1. □

The one-altered $h = 0$ boundary costs $(2 - 1/r)T$, tending to $2T$. Thus factor two is sharp *for rules required to preserve this boundary.* No minimax lower bound over freely chosen nonlinear boundaries follows.

B Reproducible evidence

The self-contained verification directory contains the following programs. Each runs with standard Python, uses exact rationals or integers, and retains explicit checks under `-O`. Counts describe finite evidence only.

Program	Principal covered claims
`verify_thresholds.py`	Capped Banzhaf probabilities by exact polynomial integration; 4,227 full profiles, 221,253 coalition checks, 16,040 own-report comparisons; four additional tail stress cases.
`verify_lower.py`	53 exact unrestricted lower certificates and 16 specified rejected corruptions.
`verify_beyond.py`	Six proportional certificates, eight corruptions, one-pool nonlinear coalition and edge checks; auxiliary archive checks.
`verify_two_pools.py`	1,330 complete labeled binary profiles, 3,129,152 coalition/profile pairs and 5,008 directed report edges.
`verify_continuous.py`	24,135 labeled profiles on fifteen grids, 3,463,360 coalition/profile pairs and 235,523 directed report edges, including zero and below-$1/r$ reports.
`verify_scope.py`	Exact substitutions for the continuous rule's many-altered-pool limitation and an alternative mixture's failure.

The command `python3 -B -O verification/run_all.py` executes these scripts in isolated output directories and verifies report/source hashes. The verification archive records its file manifest and clean replay result. These checks supplement rather than replace each proof. They do not certify novelty, rounded implementations, or the unproved unrestricted dichotomy.

References

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